

Master Thesis

Distilling Reasoning: Fine-tuning Smaller Language Models on LLM Chain-of-Thought Traces for Cryptic Crossword Solving

Rey Rashid

Supervisor Antske Fokkens
2nd reader Piek Vossen

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Vrije Universiteit Amsterdam

Computational Linguistics and Text-Mining Lab
Department of Language and Communication
Faculty of Humanities

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Abstract

This thesis investigates whether chain-of-thought reasoning trace distillation from a large language model can improve the performance and reasoning quality of a smaller model on the task of solving English cryptic crossword clues. Although teacher-generated reasoning traces have improved smaller models on tasks such as arithmetic reasoning (Ho et al., 2023), whether this approach generalizes to compositional linguistic tasks requiring precise letter-level manipulation has not been established.

Cryptic crossword solving requires identifying a definition component, classifying the wordplay mechanism, and executing the mechanism through precise letter-level operations. To investigate the transfer of these capabilities, GPT-5.4 was used as the teacher model to generate 8,863 structured five-step reasoning traces for clues drawn from a large collection of English cryptic crossword clues (Ho, 2021). GPT-4.1-mini was then fine-tuned on these traces and evaluated on a stratified test set of 496 clues against a zero-shot baseline and the teacher model. Results are reported on the full test set and a clean subset of 425 clues from which teacher content filter errors were removed.

Fine-tuning reduced exact match accuracy by 9.2 percentage points on the clean subset, with this reduction consistent across three training seeds. In contrast, structural trace quality improved substantially. The fine-tuned model achieved perfect structural completeness and answer-trace consistency, exceeding the teacher on both metrics. This discrepancy suggests the model learned to produce well-structured reasoning without acquiring the underlying reasoning capability.

Error analysis indicates that wordplay type misidentification is strongly associated with the accuracy drop. In 77.8% of regression cases, the fine-tuned model also misidentified the wordplay type, and type misidentification was associated with an incorrect answer 81.7% of the time. Although the teacher achieved 45.2% accuracy, it correctly identified the wordplay type in only 17.2% of cases. This suggests that the teacher often reached correct answers through direct execution without explicit type labelling.

These results suggest that chain-of-thought distillation does not automatically generalize to tasks in which reasoning steps are difficult to verify and teacher traces may reflect post-hoc rationalization rather than genuine solving.

Declaration of Authorship

I, Rey Rashid, declare that this thesis, titled *Distilling Reasoning: Fine-tuning Smaller Language Models on LLM Chain-of-Thought Traces for Cryptic Crossword Solving* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.

Date: August 10, 2026

Signed: Rey Rashid

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Chapter 1

Introduction

Large language models (LLMs) have shown strong capabilities on various linguistic tasks (Brown et al., 2020; Wei et al., 2022; OpenAI, 2023), but they are still challenged by more complex tasks. Methods like chain-of-thought prompting (having the model produce step-by-step reasoning before giving a final answer) have been used as an effective way of improving LLM performance on tasks that require multi-step inference (Wei et al., 2022). However, the extent to which intermediate reasoning improves performance depends on task complexity (Shojaee et al., 2025).

In cryptic crosswords, each clue contains two main components: a definition of the answer, and a wordplay that encodes it. Different mechanisms such as an anagram, a homophone, a reversal, or a hidden word may be used. An indicator word or phrase shows which wordplay type was used. The clue is designed to have a misleading surface reading which makes the task more challenging. For example, in the clue 'Pig's so crazy for dirt (6)', 'dirt' is the definition and 'Pig's so' is an anagram of the answer GOSSIP, with 'crazy' as the anagram indicator. As Efrat et al. (2021) state in their Cryptonite benchmark, solving a cryptic clue demands not only linguistic knowledge but a structured, iterative reasoning process: the solver must identify the part that functions as the definition, determine the wordplay type from the indicator, and construct the answer that satisfies both components. Rozner et al. (2021) demonstrate the limitations of earlier NLP approaches to the task, and more recently Sadallah et al. (2025) show that even state-of-the-art LLMs struggle significantly with wordplay identification and solution explanation. Solving cryptic crosswords is therefore a linguistically complex task that challenges language model reasoning.

A major motivation for this thesis is the computational and environmental costs of high-performing LLMs, as documented by Bender et al. (2021), which has led to interest in methods for transferring their capabilities to smaller, more efficient models (Zhu et al., 2024). A promising approach is knowledge distillation: training a smaller student model on the outputs of a larger teacher (Hinton et al., 2015). This approach can include the distillation of reasoning processes as well as the output. A student model can be fine-tuned on the chain-of-thought traces generated by a teacher model rather than on final answers alone (Ho et al., 2023; Shridhar et al., 2023). This thesis applies this method to cryptic crossword solving.

Specifically, GPT-5.4 is used as the teacher model to generate reasoning traces for clues from the georgeho.org dataset (Ho, 2021), a collection of English cryptic clues annotated with wordplay types. The trace format is based on the subtask decomposition by Sadallah et al. (2025): identification of the definition, naming and executing the

wordplay mechanism, verification of the final answer. The central question is whether a smaller model with weaker reasoning can be improved through knowledge distillation, and whether this improves reasoning quality as well as final answer accuracy. To investigate this, GPT-4.1-mini is fine-tuned on reasoning traces generated by the teacher model.

Most existing work on reasoning distillation evaluates student performance only based on final answer accuracy. This thesis instead uses an evaluation framework for the quality of the student's reasoning traces, including structural completeness, wordplay type correctness, and answer-trace consistency, based on the reasoning evaluation literature (Prasad et al., 2023; Tyagi et al., 2024; Hwang et al., 2025; Lee and Hockenmaier, 2025). This approach is motivated by the observation made by Lightman et al. (2024), that process-level supervision is more predictive of model performance on difficult tasks than outcome-level supervision alone, and by Barez et al. (2025)'s warning that chain-of-thought traces do not reflect a model's internal reasoning without independent verification.

The thesis makes the following contributions. First, it applies chain-of-thought reasoning distillation to cryptic crossword solving using a structured, task-based trace format derived from the subtask analysis of Sadallah et al. (2025). Second, it implements a rule-based evaluation framework for reasoning trace quality that does not use LLM-as-a-judge. Third, it reports results by wordplay type, to enable analysis of which mechanisms show the most improvement from knowledge distillation.

The following chapters are organized as follows. Chapter 2 reviews the relevant literature in five areas: cryptic crossword solving as an NLP task; knowledge distillation and reasoning transfer; chain-of-thought prompting and its theoretical foundations; supervised fine-tuning methodology; and frameworks for evaluating reasoning trace quality. Chapter 3 describes the methodology. Chapter 4 contains results. Chapter 5 discusses findings and limitations. Chapter 6 concludes.

Chapter 2

Literature Review

2.1 Cryptic Crossword Solving

Cryptic crosswords are a well-established puzzle form that have received limited attention in NLP. Figure 2.1 shows an example of such a puzzle. Efrat et al. (2021) provide a systematic formalization of the task for computational purposes by introducing the Cryptonite benchmark, a large dataset of cryptic clues drawn from *The Times* and *The Guardian*. Their work highlights that a cryptic clue is a compositional structure that requires the identification of a definition, a wordplay, and the indicator that signals the relationship between them.

Clue: *Pig's so crazy for dirt (6)*

Answer: GOSSIP
Wordplay type: anagram
Indicator: crazy
Definition: dirt

Figure 2.1: Example cryptic crossword clue with its key components

Early approaches to the task, surveyed by Rozner et al. (2021), demonstrate this difficulty. Rule-based methods used to encode the grammar of cryptic clue types performed relatively well on common patterns but had poor generalization. Neural approaches improved on accuracy but showed little evidence of actual structural understanding. These limitations motivated later LLM-based approaches, which might benefit from large amounts of text the models see during pretraining.

Sadallah et al. (2025) provide an analysis of where and why LLMs fail on the cryptic crossword task by splitting the task into three subtasks: definition extraction, wordplay type identification, and solution explanation. Models perform well on definition extraction, likely due to its resemblance to semantic tasks encountered in pretraining. Wordplay type identification and solution explanation, which require more structured reasoning are more challenging, with the performance being dependent on the wordplay type. They find that prompting models to address each subtask explicitly (chain-of-thought decomposition) leads to improvements over traditional prompting. In this thesis, I adopt a reasoning trace format informed by Sadallah et al. (2025)'s multi-step structure.

2.2 Large Language Models and Fine-Tuning

Supervised fine-tuning of pre-trained language models on task-specific instruction data is a main paradigm for adapting generic models to specialized tasks. Ouyang et al. (2022) demonstrate this with InstructGPT, which is the result of fine-tuning a pre-trained GPT-3 model on instruction-response pairs. Compared with the base model, InstructGPT's ability to produce desired outputs is substantially improved. A similar fine-tuning approach is used in this thesis, where the student model is trained on cryptic clues with the teacher's structured reasoning traces as the target outputs.

Large language models (LLMs) are neural language models that were trained on large amounts of text. They typically have the transformer architecture (Vaswani et al., 2017). Transformers use a self-attention mechanism that enables the model to rate the importance of each part of the input when generating the output tokens. This allows them to model long-range dependencies in text. Modern LLMs such as the GPT family (Brown et al., 2020; OpenAI, 2023) are autoregressive, which means they generate each token based on all previously generated tokens.

During pretraining, LLMs gain broad linguistic and world knowledge from large text corpora. This makes them strong zero-shot and few-shot reasoners in many tasks (Brown et al., 2020). However, pretraining alone does not adjust the model for specific task formats or instructions. Supervised fine-tuning (SFT) using task-specific instruction–response pairs adapts the model to a specific task or output format. Ouyang et al. (2022) showed that fine-tuning significantly improves a model's ability to generate the desired output.

In a knowledge distillation pipeline, a larger, more capable LLM is used as the *teacher* model to generate outputs that a smaller *student* model is then trained on. The teacher's reasoning traces provide a stronger supervision signal than gold answer labels alone. They not only help the student learn *what* the correct answer is, but also *how* to get to it step-by-step (Ho et al., 2023). Therefore, fine-tuning the student on these traces aims to transfer the teacher's reasoning process, rather than training on correct answers alone. The specific fine-tuning configuration is described in Chapter 3.

2.3 Chain-of-Thought Prompting and Reasoning

Wei et al. (2022) used chain-of-thought prompting to demonstrate that eliciting intermediate reasoning steps from an LLM improves performance on tasks that require multi-step inference, including arithmetic, symbolic reasoning, and commonsense inference. This finding justifies using teacher traces as training signals, as a student model that learns to produce these steps may show similar improvements after fine-tuning.

However, the effectiveness of chain-of-thought reasoning depends on task complexity. Shojaee et al. (2025) show that reasoning model performance drops on tasks above a certain complexity threshold (accuracy collapse). Cryptic crossword solving is a difficult reasoning task where chain-of-thought prompting alone may be insufficient. This thesis tests if these limits can be overcome by fine-tuning on teacher traces.

The pipeline design for this thesis is grounded in Lightman et al. (2024), who argue for process supervision over output supervision on complex reasoning tasks. In this approach, reasoning steps are evaluated and used as supervision signals for training the model, rather than only training it to produce correct final answers. They show that process supervision is more predictive of downstream performance than outcome

supervision alone. This finding motivates trace quality evaluation in this thesis to evaluate the effect of distillation.

A criticism of chain-of-thought reasoning is made by Barez et al. (2025), who argue that chain-of-thought traces cannot meaningfully explain a model's reasoning. They identify two failure modes: prompt-induced biases in the trace, and silent error correction, where corrections are made without being reflected in the reasoning. These findings motivate filtering teacher traces for answer-reasoning consistency before fine-tuning, and verifying trace quality using rule-based metrics.

2.4 Knowledge Distillation and Reasoning Transfer

The concept of knowledge distillation was introduced by Hinton et al. (2015) as a method for transferring the knowledge of a large, high-performing teacher model to a smaller student model. In their approach, the teacher is first trained on a standard classification task, such as image classification on MNIST or acoustic modelling for speech recognition. Instead of training the student only on the hard labels of the data (e.g. "class 3"), the student is trained to match the teacher's softened output probability distribution. The main idea is that the teacher's distributions encode not just which class is correct, but also how the teacher distinguishes between classes, which can improve the student's generalization ability. This insight was later developed into reasoning trace distillation, where the student is trained on full reasoning sequences rather than just final answers.

As LLMs are continuously increasing in scale and cost, the interest in distillation has grown. Zhu et al. (2024) survey model compression methods for large language models, discussing inference time, latency, and resource challenges posed by these models. Knowledge distillation is presented as one of the main approaches for improving efficiency and deployability.

The extension of distillation to include not only the output but also the reasoning processes is explored in more recent work. Ho et al. (2023) demonstrate that fine-tuning smaller models on the chain-of-thought traces generated by a large teacher model transfers reasoning ability. Students trained on traces outperform both their untuned baselines and models trained on final-answer supervision alone, suggesting that the reasoning traces contain knowledge that can be transferred.

Shridhar et al. (2023) show that reasoning capabilities can be distilled through problem decomposition (Socratic CoT). Later work confirms that trace distillation gains depend on the teacher-student capacity gap (Zhang et al., 2025). When the student does not have the representational capacity to absorb the teacher's reasoning, distillation may produce superficial trace imitation rather than actual knowledge transfer.

Dai et al. (2025) propose identifying and emphasizing the key reasoning steps in teacher traces during distillation, showing that some steps contribute more to generalization. Li et al. (2025) introduce a peer-review mechanism in which multiple teacher models evaluate and filter each other's traces before fine-tuning. These works are the motivation for designing a five-step trace format in which reasoning steps are explicit and individually verifiable, and for implementing quality filtering on generated traces.

Zelikman et al. (2022) propose STaR, a self-improvement approach in which a model iteratively generates its own reasoning traces and fine-tunes on those with correct final answers. In this approach, the model's own reasoning acts as a training signal instead of teacher supervision. It therefore requires the student to already be capable of producing

correct traces, which is difficult for a challenging task like cryptic crosswords. This thesis uses an external teacher rather than self-generated traces to ensure a more reliable supervision signal.

2.5 Research Question, Hypothesis, and Contribution

Upon reviewing the literature on cryptic crossword solving, reasoning trace distillation, and reasoning trace evaluation, this thesis aims to answer the following research question:

> *To what extent does reasoning trace distillation from a large language model improve the performance and reasoning quality of a smaller, weaker model on the task of solving English cryptic crossword clues?*

Based on the findings reviewed in this chapter, the following is hypothesized. Fine-tuning on structured teacher traces outperforms zero-shot prompting on cryptic crossword accuracy, consistent with Ho et al. (2023). However, given the difficulty of cryptic crossword solving and the capacity gap between teacher and student, improvements are expected to vary based on wordplay types, with structurally simpler types (e.g. hidden words, anagrams) showing more improvement. Additionally, based on Lightman et al. (2024) and Barez et al. (2025), improvements in answer accuracy may not fully correspond to improvements in reasoning trace quality. The student model may generate some correct answers with structurally incomplete or inconsistent traces.

The thesis addresses this by implementing a multi-dimensional trace evaluation framework based on the reasoning evaluation literature, which provides a more detailed insight into what the student model has learned through distillation than what final-answer accuracy alone can provide.

Chapter 3

Methodology

This chapter describes the data collection and processing (Section 3.2), the models (Section 3.3), the baseline evaluation (Section 3.4), the trace generation process (Section 3.5), the fine-tuning setup (Section 3.6), and the evaluation framework (Section 3.7).

3.1 Overview

The overall methodology is knowledge distillation in which a large, more capable teacher model (GPT-5.4) is used to generate structured reasoning traces for cryptic crossword clues. These traces are then used to fine-tune a smaller student model (GPT-4.1-mini). The pipeline has five stages: (1) data collection and preprocessing; (2) dataset splitting; (3) baseline evaluation; (4) teacher trace generation; and (5) fine-tuning. Each is described below.

3.2 Data

3.2.1 Source Data

The data used in this thesis comes from the georgeho.org cryptic crossword dataset (Ho, 2021). The dataset provides two source files: a large collection of cryptic crossword clues with gold answers (660,613 entries), and a set of indicator annotations (15,735 entries) that links each clue to its wordplay type and the indicator word or phrase that signals that type.

3.2.2 Preprocessing

Processing the raw data has several steps. First, each indicator entry, which may reference multiple clue IDs, is expanded so that each indicator–clue pair is in a single row, resulting in 94,274 rows. These are then matched to their corresponding clues to produce a single dataset of annotated clues.

Two filters are applied to ensure data quality and simplify evaluation. Clues with multi-word answers are removed, as they introduce structural complexity that goes beyond the scope of this study. This removes 16,443 clues. Additionally, rows in which the indicator string does not appear in the clue text are removed, leaving a further

13,412 rows. After dropping 14 rows with null values in any required column, the final clean dataset has 62,918 clues. Table 3.1 shows the distribution of wordplay types.

Wordplay type	**Count**	**%**
Anagram	28,993	46.1
Container	9,324	14.8
Reversal	9,184	14.6
Insertion	7,572	12.0
Homophone	3,219	5.1
Hidden	2,575	4.1
Deletion	1,453	2.3
Alternation	598	0.9
Total	**62,918**	**100.0**

Table 3.1: Wordplay label distribution in the clean dataset.

3.2.3 Dataset Splits

The dataset is divided into three splits. A stratified test set of 496 clues is held out by sampling 62 clues per wordplay type. Since the test set is balanced across wordplay types, performance can be compared per category without bias from class imbalance. The remaining 62,422 clues are split 80/20 into a training set (49,937 clues) and a development set (12,485 clues), stratified by wordplay type and shuffled with a fixed random seed for reproducibility. The training and development splits are not stratified and maintain the natural corpus distribution. Table 3.2 shows the resulting split sizes by wordplay type.

Wordplay type	**Train**	**Dev**	**Test**
Anagram	23,201	5,730	62
Container	7,417	1,845	62
Reversal	7,229	1,893	62
Insertion	6,041	1,469	62
Homophone	2,516	641	62
Hidden	2,002	511	62
Deletion	1,097	294	62
Alternation	434	102	62
Total	**49,937**	**12,485**	**496**

Table 3.2: Dataset split sizes by wordplay type.

3.3 Models

3.3.1 Teacher Model

GPT-5.4 is used as the teacher model, accessed through Microsoft Azure (OpenAI, 2023). It is capable of producing structured, coherent five-step reasoning traces when

prompted with a specific output format. GPT-5.4 is used only for trace generation and baseline evaluation and is not fine-tuned.
A challenge encountered during this project was that GPT-5.4 has a stricter content filtering policy than the student model. A number of clues triggered content filter errors during both baseline evaluation and trace generation. These clues were skipped and replaced. Section 3.4 provides further detail on how these errors were handled.

3.3.2 Student Model Candidates

The two student model candidates are GPT-4.1-mini and GPT-4.1-nano (OpenAI, 2023). Both are members of the GPT-4.1 model family. GPT-4.1-mini offers a balance of capability and efficiency, and GPT-4.1-nano is the smallest variant. A baseline evaluation (Section 3.4) is used to select the better candidate for fine-tuning.

3.4 Baseline Evaluation

Before fine-tuning, a zero-shot baseline experiment is done on all three models to see the performance gap between teacher and student candidates. Each model is prompted on 100 clues from the training set. The prompt used is shown in Figure 3.1. It consists of a system message that describes the structure of a cryptic clue and the eight wordplay types, and a user message that contains the clue and instructions to go through five steps: identifying the definition, determining the wordplay type, identifying the indicator, executing the wordplay, and verifying the answer against the definition.

Clues that trigger content-filter errors for any of the three models are skipped and replaced with the remaining training examples until 100 valid results are found. This ensures that all three models are evaluated on exactly the same 100 clues. Temperature is set to 0.0 for reproducibility.

System message (all models, all stages):

You are an expert cryptic crossword solver with deep knowledge of all wordplay mechanisms. Your task is to solve cryptic clues by reasoning through them in a precise, structured way.

A cryptic clue always has exactly two parts:
- A DEFINITION: a synonym, hyponym, or description of the answer (usually at the start or end of the clue)
- A WORDPLAY component: encodes the answer through a mechanism such as:
  * Anagram (indicator words: mixed, confused, broken, scrambled, etc.)
  * Container (one word holds another: in, within, holding, around, etc.)
  * Reversal (back, returned, going up, reflected, etc.)
  * Insertion (into, entering, inside, etc.)
  * Homophone (sounds like, heard, say, spoken, etc.)
  * Hidden (in, within, some of, part of, etc.)
  * Deletion (without, losing, drop, remove, etc.)
  * Alternation (every other, odd letters, even letters, etc.)

Always follow the structured format exactly.

User message (baseline and evaluation):

Cryptic clue: *[clue text]*

Work through this step by step:
Step 1 - Identify the definition component
Step 2 - Identify the wordplay type (must be one of: anagram, container, reversal, insertion, homophone, hidden, deletion, alternation)
Step 3 - Identify the wordplay indicator
Step 4 - Execute the wordplay
Step 5 - Verify the answer matches the definition

End your response with exactly: `ANSWER: [your answer in capitals]`

Figure 3.1: System and user messages used for baseline evaluation and final evaluation. The system message is identical across all models and all pipeline stages. The user message uses the same five-step structure as the fine-tuning training examples.

The results show a significant performance gap between the teacher and student models. GPT-5.4 has 47.0% accuracy on the 100-clue sample, GPT-4.1-mini 25.0%, and GPT-4.1-nano 11.0%. Accuracy per wordplay type (Table 3.3) shows that the gap is largest for reversal and alternation, where both students score 0%. Per-type figures should however be interpreted with caution as the sampling was random.

Wordplay type	**GPT-5.4**	**GPT-4.1-mini**	**GPT-4.1-nano**
Hidden	66.7%	0.0%	0.0%
Anagram	58.0%	34.0%	18.0%
Alternation	50.0%	0.0%	0.0%
Deletion	50.0%	0.0%	50.0%
Homophone	44.4%	22.2%	11.1%
Container	41.7%	16.7%	0.0%
Insertion	25.0%	25.0%	0.0%
Reversal	9.1%	18.2%	0.0%
Overall	**47.0%**	**25.0%**	**11.0%**

Table 3.3: Zero-shot baseline accuracy by wordplay type

Based on these results, GPT-4.1-mini is selected as the student model. Its 25.0% accuracy shows a stronger starting point for distillation compared to GPT-4.1-nano's 11.0%. GPT-4.1-nano's near-zero performance across most wordplay types suggests it may lack the capacity to learn reasoning from teacher traces.

3.5 Trace Generation

3.5.1 Exploratory Phase and Quality Filtering

In an exploratory phase, 1,000 traces were generated to determine which quality filters should be applied. Three candidate filters were tested: (1) a *structural completeness filter* that checks if all five step labels are present in the trace; (2) a *wordplay-type match filter* to verify that the wordplay type named in Step 2 matches the gold label by searching for the type name or common synonyms; and (3) an *indicator match filter* to verify that the indicator identified in Step 3 matches the gold annotation.

After looking at a small sample of traces rejected by the indicator filter, it seemed that the filter was overly strict. In many cases, the teacher correctly identified a substring of the gold indicator phrase. For example, identifying *abandons* when the gold label was *church abandons*, or *to* when the gold was *contribution to*. Since these cases reflect correct reasoning rather than errors, the indicator filter was removed to avoid discarding useful traces.

Inspection of traces rejected by the wordplay-type filter revealed a different issue. A large number of rejections reflected the genuine difficulty of this sub-task, and the ambiguity between some wordplay types. In particular, insertion clues were often mislabeled as container clues, and hidden clues were mislabeled as anagram clues. These confusions are expected given that insertion and container both involve one substring being placed inside another, and that some hidden clues resemble anagram clues on the surface. The 1,000-trace exploratory run led to 398 rejections from the wordplay-type filter, which is a high rejection rate that would discard a large portion of usable data.

As a result, the wordplay-type filter was removed. Instead, the gold wordplay type was added to the trace generation prompt (Figure 3.2) to ensure that the teacher always has the correct type before generating its explanation. By keeping only the structural completeness filter, the acceptance rate on the exploratory run reached 74.7%.

3.5.2 Design and Procedure

Structured chain-of-thought (CoT) reasoning traces are generated using GPT-5.4 as the teacher. Unlike the zero-shot baseline experiment, where the model is provided only with the clue, trace generation is *answer-informed*: the teacher is given the clue, the gold answer, and the gold wordplay type, and is asked to explain the reasoning using a five-step format (Figure 3.2). This is meant to produce high-quality reasoning traces that will be reliable enough for supervised fine-tuning, and prevent the teacher from rationalizing an incorrect answer in its reasoning. Temperature is set to 0.7 for trace generation to allow some diversity in expression for the student to learn from, rather than repetitions of a single phrasing.

Based on the exploratory phase, the final pipeline retains three quality filters. The primary filter is a structural completeness check that verifies all five step labels are present. A wordplay-type match check and an answer correctness check are used as safety checks as both the gold wordplay type and the gold answer are explicitly provided

in the trace generation prompt. Additionally, clues that trigger content filter errors from GPT-5.4 are also excluded from fine-tuning.

Traces are generated separately for the training and development splits. A target of 1,250 accepted traces per wordplay type is set (`N_TRACES = 10000`, giving `per_label = 1250`). For the six common wordplay types (anagram, container, reversal, insertion, homophone, and hidden) the full 1,250-trace requirement is met. However, the rarest types are limited by the smaller number of available clues in the training set. This results in 967 accepted traces for deletion and 396 for alternation. The final training set of 8,863 traces is therefore not perfectly stratified. The overall acceptance rate is 87.0%, with 1,319 traces rejected.

For the development split, a target of 250 traces per type (`n_traces = 2000`) is set. Common types and deletion reach 250, but alternation runs out at 93 traces. The development set has 1,843 traces in total.

User message (trace generation):

Cryptic clue: *[clue]*
Correct answer: *[answer]*
Wordplay type: *[wordplay]*

Explain how to solve this clue using the exact format below. Be precise and educational — your explanation will be used to teach a smaller model how to reason about cryptic clues.

Work through this step by step:
Step 1 - Identify the definition component
Step 2 - Identify the wordplay type (must be one of: anagram, container, reversal, insertion, homophone, hidden, deletion, alternation)
Step 3 - Identify the wordplay indicator
Step 4 - Execute the wordplay
Step 5 - Verify the answer matches the definition

End your response with exactly: `ANSWER: [your answer in capitals]`

Figure 3.2: User message used for teacher trace generation. The five-step format is identical to the evaluation and fine-tuning prompts to ensure consistency in all stages.

3.6 Fine-Tuning

3.6.1 Data Preparation

Accepted traces are converted into the JSONL format required for supervised fine-tuning on Azure AI Foundry. Each training example is formatted as a chat completion object with three parts: (1) a system message with the task description and wordplay type definitions; (2) a user message containing the clue and five-step instructions; and (3) an assistant message containing the teacher's full five-step reasoning trace.

The user message used during fine-tuning is identical to the user message used during evaluation (Figure 3.1). This ensures prompt consistency between training and evaluation.

3.6.2 Training Configuration

Hyperparameters were selected based on a manual experiment on a 2000-trace training set, comparing two learning rate multipliers while keeping all other settings fixed. A

more exhaustive hyperparameter search was not conducted due to the computational cost of fine-tuning. Two learning rate multipliers (1.0 and 2.0) were trained for 5 epochs. For both settings, validation performance peaked at the end of epoch 3, with the lower multiplier (1.0) leading to better results. A smaller learning rate multiplier of 0.5 was therefore selected for the final training configuration as a conservative choice expected to further improve generalization and reduce overfitting risk. However, this was not empirically verified against 1.0 due to the same cost constraints.

Since Azure AI Foundry keeps only the three most recent checkpoints during fine-tuning, final training was run for 4 epochs so that the saved checkpoints cover epochs 2, 3, and 4 to see changes in validation performance. The final fine-tuning configuration therefore uses a learning rate multiplier of 0.5, 4 epochs, and a batch size automatically set by Azure AI Foundry.

For the hyperparameter selection experiment, the random seed was fixed at 42 to ensure reproducibility. To assess the stability of fine-tuning results, the final configuration is trained using three different random seeds (0, 42, and 100). This allows us to observe variation due to training stochasticity.

3.7 Evaluation Framework

3.7.1 Models Evaluated

The evaluation compares six models on the full 496-clue test set:

1. **Baseline (mini)**: GPT-4.1-mini, zero-shot (lower bound)
2. **Baseline (nano)**: GPT-4.1-nano, zero-shot (additional reference)
3. **Fine-tuned seed 1**: GPT-4.1-mini fine-tuned, seed 1
4. **Fine-tuned seed 42**: GPT-4.1-mini fine-tuned, seed 42
5. **Fine-tuned seed 100**: GPT-4.1-mini fine-tuned, seed 100
6. **Teacher**: GPT-5.4, zero-shot (upper bound)

All models are evaluated using the same prompt (Figure 3.1) for fair comparison. Fine-tuned model results are reported as the mean ± standard deviation across the three seeds to show training stability.

3.7.2 Practical Challenges

Reasoning loops and token limits: Smaller models, particularly GPT-4.1-mini, sometimes got stuck in reasoning loops and produced verbose responses that did not end cleanly in the expected format, leading to responses cut off before the `ANSWER:` line. Increasing the token limit from 600 to 2000 reduced truncation while preventing excessively long responses.

Answer format variation: Models occasionally varied the format of their answer rather than using the exact `ANSWER: WORD` format. To handle this format variation, six fallbacks were added to the function that extracts answers from model responses. The primary pattern searches for the explicit `ANSWER: WORD` string. Subsequent fallbacks handle variations such as bolded answers (`**WORD**`), all-caps words on their own line, and inline answer statements (e.g. *the answer is*).

Structured outputs attempt: Azure's structured outputs API (via Pydantic schema with `.parse()`) were attempted to guarantee clean answer extraction and remove the need for regex-based parsing. Structured outputs constrain the wordplay type field to exactly the eight valid types, removing the need for some filtering steps. However, this introduced a new problem: GPT-4.1-mini produced extremely verbose responses when filling structured JSON fields, consistently hitting the token limit before completing the JSON object, causing a parse failure even at very high limits like 4,000 tokens. As a result, the pipeline was adjusted to free-form text approach again.

Content filter errors: GPT-5.4 has a stricter content filtering policy than the student models, leading to `invalid_prompt` errors during trace generation and evaluation. Content filter errors during the final evaluation are handled differently from previous stages. Instead of replacing errored clues, they are recorded with an empty prediction (counted as incorrect) for that model and kept in the results file.

Two versions of results are reported: a full version (496 clues, content filter errors counted as incorrect) and a clean subset version. For the clean subset, clues that triggered content filter errors for the teacher model (GPT-5.4) are removed from all model results before computing summary statistics, as they only reflect the teacher's stricter content policy. Empty predictions from baseline and fine-tuned models are kept and counted as incorrect, as these represent extraction failures rather than API-level errors.

3.7.3 Primary Metric: Exact Match Accuracy

The primary evaluation metric is exact-match accuracy: a prediction is counted as correct if the predicted answer string matches the gold answer exactly, after uppercasing and stripping whitespace from both strings. This is a strict metric where partial matches are not credited.

Per-type accuracy is reported alongside overall accuracy to enable analysis of performance variation across the eight wordplay types, which differ substantially in structural complexity.

3.7.4 Background on Trace Evaluation

Since correct final answers can be reached through incorrect reasoning, evaluation of reasoning traces provides more information than final-answer accuracy alone. Lee and Hockenmaier (2025) survey approaches to step-by-step reasoning trace evaluation. They identify four main criteria: correctness (factual validity of steps), coherence (steps forming a connected argument), relevance (steps contributing toward the final answer), and completeness (presence of all steps). These criteria inform the rule-based evaluation framework in this thesis. Prasad et al. (2023) define correctness in their ReCEval framework as steps being both accurate and informative. The answer-trace consistency metric in this thesis is inspired by this framework: it checks whether the final answer is supported by the reasoning in Steps 4 and 5 of the trace, addressing the silent error correction failure mode identified by Barez et al. (2025).

Hwang et al. (2025) propose principal knowledge grounding, evaluating reasoning steps based on task-specific atomic knowledge items using precision and recall. The wordplay type correctness metric in this thesis is informed by this approach: Step 2 of each trace is evaluated against the wordplay taxonomy of Sadallah et al. (2025),

the core atomic knowledge for this task. Tyagi et al. (2024) propose PuzzleEval's error taxonomy for evaluating grid puzzle reasoning chains, using error type, locus, and severity. Like cryptic crossword solving, grid puzzles require compositional reasoning where errors propagate forward across steps. This research informs the step-level structural completeness scoring in this thesis. Lee et al. (2025) propose CheckEval, a binary question-answering checklist approach using LLM-as-a-judge. The checklist structure inspires the decomposition of trace quality into separate dimensions in this thesis. However, I use rule-based and string-matching methods rather than LLM-as-a-judge to ensure fully deterministic and reproducible evaluation.

These frameworks jointly motivate the three metrics defined in the following subsection.

3.7.5 Reasoning Trace Quality Metrics

Three complementary rule-based metrics are used to evaluate the quality of reasoning traces, motivated by the argument of Lightman et al. (2024) that process-level evaluation is more informative than outcome-level evaluation alone for difficult reasoning tasks.

Structural completeness (0–5): the number of defined step labels (Step 1-Step 5) present in the trace. A score of 5 indicates all steps are present; lower scores indicate incomplete traces. This metric is based on the completeness criterion of Lee and Hockenmaier (2025) and the structural analysis of Tyagi et al. (2024).

Wordplay type correctness (0 or 1): whether the wordplay type named in Step 2 matches the gold label. Matching checks whether the gold wordplay type name appears as a substring in the Step 2 text of the trace. This metric is motivated by Hwang et al. (2025)'s principal knowledge grounding framework and the taxonomy of Sadallah et al. (2025).

Answer-trace consistency (0 or 1): whether the predicted answer appears in Steps 4 or 5 of the trace, checking whether the answer is argued for within the reasoning rather than simply appended at the end. This metric is inspired by the correctness dimension of ReCEval (Prasad et al., 2023) and addresses the silent error correction concern of Barez et al. (2025).

An aggregate trace quality score is computed as the mean of available normalized scores (structural completeness divided by 5, wordplay type correctness, answer-trace consistency), providing a single summary value for comparison. Together, these three metrics provide a more informative picture of what the student model has learned during fine-tuning than final-answer accuracy alone.

Chapter 4

Results

This chapter presents the results of evaluating all models on the test set: the GPT-4.1-mini baseline, the fine-tuned GPT-4.1-mini student (with three different seeds), and the GPT-5.4 teacher. Results are reported in two versions: a full test set version (n=496, stratified at 62 clues per wordplay type) in which the 71 clues that triggered content filter errors for the teacher model are counted as incorrect, and a clean subset version (n=425) in which those clues are removed from all model results to ensure a fair comparison. The results on the clean subset are treated as primary unless otherwise stated, as they allow for a comparison that is not impacted by content filtering errors.

Results are organized as follows. Section 4.1 presents exact match accuracy for all models. Section 4.2 reports accuracy based on wordplay type. Section 4.3 presents reasoning trace quality scores. Section 4.4 relates the findings to the research question.

4.1 Exact Match Accuracy

Table 4.1 summarises overall exact match accuracy for all models on both the full test set and the clean subset. The results for fine-tuned models are reported as the mean and standard deviation in the three different seeds to show training stability. Only GPT-4.1-mini was selected for fine-tuning based on the baseline evaluation results (Section 3.4). GPT-4.1-nano is included as an additional reference point.

Model	**Full (n=496)**	**Clean (n=425)**
Baseline GPT-4.1-mini	23.8%	26.6%
Baseline GPT-4.1-nano	10.9%	12.5%
Fine-tuned GPT-4.1-mini seed 1	15.3%	17.2%
Fine-tuned GPT-4.1-mini seed 42	15.5%	17.7%
Fine-tuned GPT-4.1-mini seed 100	15.1%	17.2%
Fine-tuned (mean ± std)	**15.4% ± 0.2%**	**17.4% ± 0.3%**
Teacher (GPT-5.4)	38.7%	45.2%

Table 4.1: Exact match accuracy on the full test set (n=496, content filter errors counted as incorrect) and the clean subset (n=425, teacher errors removed from all models). Fine-tuned results are reported as mean ± standard deviation across three seeds.

Several observations can be made from Table 4.1. First, the teacher model significantly outperforms the student model, with 45.2% accuracy on the clean subset compared to 26.6% for the baseline and 17.4% for the fine-tuned student. This confirms that GPT-5.4 has much stronger zero-shot reasoning ability on cryptic crosswords, which is consistent with the baseline evaluation results reported in Chapter 3.

Second, the fine-tuned model performs *worse* than the baseline model on both the full test set (15.4% vs 23.8%) and the clean subset (17.4% vs 26.6%). Fine-tuning on the teacher's reasoning traces did not improve final-answer accuracy and instead decreased it compared to the baseline model. This is the main finding of this thesis and will be examined in detail in Chapter 5.

Third, fine-tuning stability is very high. The standard deviation in different seeds is 0.2–0.3 percentage points, which shows that training stochasticity is not a meaningful source of variance. The drop in performance compared to the baseline model is consistent in all three seeds. This rules out the possibility of the result being specific to a particular training run.

The gap between the results of the full and clean test sets is remarkably larger for the teacher (38.7% vs 45.2%) than for the other models. This is a reflection of the 71 content filter errors that disproportionately penalize GPT-5.4 because each error is counted as incorrect in the full results.

Of the 71 clues removed from the clean subset due to teacher content filter errors, only 5 (7.0%) were answered correctly by the baseline and 2 (2.8%) by the fine-tuned model, compared to overall accuracies of 23.8% and 15.4% respectively. This suggests that the removed clues were genuinely harder than average for all models, and that their removal does not artificially inflate the clean subset results for the student models.

4.2 Accuracy by Wordplay Type

Table 4.2 shows exact match accuracy broken down by wordplay type for the clean subset. Fine-tuned accuracy is the mean across three seeds (1, 42, 100).

Wordplay type	**Baseline**	**Fine-tuned**	**Teacher**	**Δ FT**
Alternation	46.6%	20.7%	36.2%	−25.9%
Hidden	35.2%	13.6%	55.6%	−21.6%
Homophone	20.7%	11.5%	43.1%	−9.2%
Anagram	33.3%	25.9%	57.4%	−7.4%
Deletion	17.0%	12.6%	32.1%	−4.4%
Insertion	18.8%	13.2%	43.8%	−5.6%
Container	25.5%	25.5%	55.3%	±0.0%
Reversal	13.2%	16.4%	39.6%	+3.1%
Overall	**26.6%**	**17.4%**	**45.2%**	−9.2%

Table 4.2: Exact match accuracy by wordplay type on the clean subset (n=425). Fine-tuned accuracy is the mean across the three seeds. Δ FT shows the change from baseline to fine-tuned mean, sorted by the amount of regression.

The per-type results show that the drop in performance after fine-tuning is not the same in all wordplay types. Six out of eight wordplay types show regression compared

to the baseline, with alternation (−25.9%) and hidden (−21.6%) showing the largest declines in performance. Container shows no change (±0.0%) and reversal shows a small improvement (+3.1%).

The largest regressions occur for alternation and hidden, which are the same types where the baseline model performed relatively well zero-shot (46.6% and 35.2% respectively). This suggests that the baseline model might use surface-level strategies that are effective for these types. Alternation answers can sometimes be reached by scanning for regular letter patterns, and hidden answers are literally present in the clue text. Fine-tuning seems to disrupt these strategies without replacing them with better reasoning.

The teacher shows the largest advantages over baseline on anagram (+24.1%), container (+29.8%), insertion (+25.0%), and hidden (+20.4%). These are wordplay types that require precise letter-level manipulation. This is consistent with Sadallah et al. (2025)'s finding that wordplay execution is one of the most challenging subtasks for LLMs, and suggests that the teacher's better performance is driven by stronger compositional reasoning.

One anomaly is worth noting. The teacher performs worse than the zero-shot baseline on alternation (36.2% vs 46.6%). This may be due to sampling effects given the small per-type test set size (62 clues per type before error removal), which makes per-type estimates sensitive to individual clue difficulty.

4.3 Reasoning Trace Quality

Table 4.3 shows reasoning trace quality scores for all models on the clean subset.

Model	**Structural comp. (0–5)**	**Answer consistency**	**Wordplay correctness**	**Aggregate (0–1)**
Baseline GPT-4.1-mini	4.41	0.51	0.41	0.60
Baseline GPT-4.1-nano	4.70	0.70	0.49	0.71
Fine-tuned GPT-4.1-mini (mean)	5.00	1.00	0.24	0.75
Teacher (GPT-5.4)	4.99	0.98	0.17	0.72

Table 4.3: Reasoning trace quality scores on the clean subset (n=425). Structural completeness is scored 0–5; answer consistency and wordplay correctness are binary (0 or 1), reported as proportions. Fine-tuned scores are the mean across the three seeds.

The trace quality results reveal a clear difference between accuracy and trace quality. The fine-tuned model achieves the highest scores on structural completeness (5.00/5) and answer-trace consistency (1.00), exceeding the teacher on both criteria, although it has the lowest accuracy of all models except for GPT-4.1-nano.

Structural completeness. The fine-tuned model produces perfectly structured five-step traces on nearly all clues (5.00/5), and significantly exceeds the baseline (4.41/5). This shows that fine-tuning successfully transferred the step-by-step format from the teacher's reasoning traces. The baseline's lower score reflects its tendency to occasionally skip steps or produce truncated reasoning.

Answer-trace consistency. The fine-tuned model scores 1.00 on answer-trace consistency: the predicted answer almost always appears within steps 4 or 5 of the trace. The baseline scores only 0.51. This suggests a genuine structural improvement:

the fine-tuned model has learned to argue for its answer within the reasoning body rather than appending it separately. However, since many of these answers are wrong, the metric primarily shows correct formatting rather than valid reasoning.

Wordplay type correctness. The fine-tuned model scores much lower on wordplay type correctness (0.24) than both the GPT-4.1-mini baseline (0.41) and the GPT-4.1-nano baseline (0.49). This is the only trace quality metric that becomes worse after fine-tuning, and will be examined further in the error analysis in Section 4.5

4.4 Relation to the Research Question

The research question of this thesis is:

> *To what extent does reasoning trace distillation from a large language model improve the performance and reasoning quality of a smaller, weaker model on the task of solving English cryptic crossword clues?*

The answer suggested by the results is mainly negative for accuracy and mixed for trace quality.

Fine-tuning on the teacher's reasoning traces did not improve exact match accuracy. The fine-tuned student model scores 17.4% on the clean subset, compared to 26.6% for the zero-shot baseline, which is a regression of 9.2%. This is consistent in all three training seeds and six out of eight wordplay types. The hypothesis that fine-tuning would outperform zero-shot prompting, consistent with Ho et al. (2023), is not supported.

The secondary hypothesis that improvements would be different for each wordplay type is partially supported. Reversal is the only wordplay type that shows a small improvement over the baseline (16.4% vs. 13.2%). Container is unchanged. All other types show regression, with alternation and hidden showing the largest drops.

The hypothesis regarding the difference between accuracy and trace quality is clearly supported by the results. The fine-tuned model significantly improves in structural completeness and answer-trace consistency, but worsens in accuracy and wordplay correctness. This pattern is consistent with Lightman et al. (2024)'s argument that process-level and outcome-level performance can differ. It also supports the concern identified by Barez et al. (2025) that chain-of-thought traces may reflect format imitation rather than actual reasoning.

4.5 Error Analysis

To investigate the causes of the accuracy drop after fine-tuning, this section examines correlations between the evaluation metrics and reports on the overlap between correct predictions across models. All analyses use the clean subset (n=425).

4.5.1 Wordplay Type Correctness and Answer Correctness

Table 4.4 shows the joint distribution of wordplay type correctness and answer correctness for the baseline, fine-tuned, and teacher models.

	Type correct		Type wrong	
Model	**Answer correct**	**Answer wrong**	**Answer correct**	**Answer wrong**
Baseline (GPT-4.1-mini)	42 (9.9%)	134 (31.5%)	71 (16.7%)	178 (41.9%)
Fine-tuned (seed 42)	16 (3.8%)	87 (20.5%)	59 (13.9%)	263 (61.9%)
Teacher (GPT-5.4)	27 (6.4%)	46 (10.8%)	165 (38.8%)	187 (44.0%)

Table 4.4: Joint distribution of wordplay type correctness and answer correctness on the clean subset (n=425).

Model	**Correct type → correct answer**	**Wrong type → wrong answer**
Baseline (GPT-4.1-mini)	23.9%	71.5%
Fine-tuned (seed 42)	15.5%	81.7%
Teacher (GPT-5.4)	37.0%	53.1%

Table 4.5: Conditional accuracy rates by wordplay type correctness on the clean subset (n=425).

Several observations can be made from Tables 4.4 and 4.5. Firstly, wrong wordplay type identification is a strong indicator of a wrong answer for all models, but especially for the fine-tuned model. Of the 322 clues where the fine-tuned model identified the wrong wordplay type, 263 (81.7%) also had a wrong answer, compared to 178 out of 249 (71.5%) for the baseline and 187 out of 352 (53.1%) for the teacher (Table 4.5). However, even correct type identification does not guarantee a correct answer. When the fine-tuned model identifies the type correctly, it still only produces the right answer 15.5% of the time, compared to 23.9% for the baseline and 37.0% for the teacher (Table 4.5). This confirms that correct type identification is necessary but not sufficient for a correct answer.

Secondly, the fine-tuned model produces fewer correct type identifications than the baseline (103 vs. 176 out of 425). Combined with the higher wrong-type-to-wrong-answer rate (81.7% vs. 71.5%), this suggests that wrong type identification is more damaging for the fine-tuned model than for the baseline, as the fine-tuned model recovers less often from a wrong type identification than the baseline does.

The teacher's pattern is distinctly different. It frequently reaches the correct answer despite its low wordplay type correctness, with 165 out of 425 clues (38.8%) having an incorrect type-correctness score but a correct answer. This indicates that the teacher's ability to reach the correct answer is less dependent on correct type identification. In contrast, correct type identification and correct answer are much more closely linked for the fine-tuned model. This suggests that fine-tuning on traces where Step 2 always named the correct type may have introduced a dependence on this step, which constrains the model's ability to recover from type misidentification.

4.5.2 Teacher Wordplay Type Correctness

A specific motivation for this analysis was to determine whether the teacher's low wordplay type correctness score (17.2%) reflects genuine type-identification errors or a formatting mismatch between the model's output and the substring matching metric. The analysis confirmed that the gold wordplay type name was absent from Step 2 text in all 352 cases where the teacher scored 0.

Examining the Step 2 text in these cases shows that in the majority of cases (309 out of 352, 87.8%), the teacher skips naming the type entirely and instead begins executing the mechanism directly (e.g. taking alternating letters from the clue words). In 43 cases (12.2%), it identifies the wrong type (e.g. classifying an alternation clue as a homophone). The teacher's high accuracy (45.2%) despite low type correctness (17.2%) suggests it can reach the correct answer through execution even when it skips or misidentifies the wordplay type.

4.5.3 Wordplay Type Correctness by Type

Table 4.6 shows wordplay type correctness rates by wordplay type for the baseline and fine-tuned model, alongside accuracy.

Type	Baseline accuracy	FT accuracy	Baseline wordplay	FT wordplay
Anagram	33.3%	22.2%	74.1%	24.1%
Homophone	20.7%	13.8%	67.2%	15.5%
Container	25.5%	21.3%	53.2%	12.8%
Hidden	35.2%	13.0%	51.9%	42.6%
Reversal	13.2%	22.6%	47.2%	34.0%
Deletion	17.0%	13.2%	15.1%	43.4%
Insertion	18.8%	14.6%	10.4%	4.2%
Alternation	46.6%	20.7%	10.3%	15.5%

Table 4.6: Accuracy and wordplay type correctness by wordplay type for baseline and fine-tuned (seed 42) models on the clean subset.

The most notable finding is the substantial drop in anagram wordplay type correctness after fine-tuning (from 74.1% to 24.1%). Since anagram is among the most frequent types in the training data (1,250 traces), this is unexpected. One possible explanation is that the model may have overfitted to surface features of frequent wordplay types, making type identification less generalizable to unseen test clues. Anagram, homophone, and container show the largest drops (50, 52, and 40 percentage points respectively). However, this pattern is not consistent across all abundant types. Reversal, insertion, and hidden show much smaller drops despite also having 1,250 training traces. The explanation therefore remains speculative.

Three types show a discrepancy between type correctness and accuracy. Deletion type correctness nearly triples (15.1% to 43.4%) yet accuracy decreases (17.0% to 13.2%). Alternation type correctness also improves slightly (10.3% to 15.5%) while accuracy drops significantly (46.6% to 20.7%). Reversal shows the opposite pattern: type correctness drops (47.2% to 34.0%) while accuracy slightly improves (13.2% to 16.4%). These cases suggest that correct type identification does not guarantee correct answers, and incorrect type identification does not always prevent them.

Alternation and insertion show very low type correctness for both baseline and fine-tuned models (both below 16%), which suggests that these mechanisms are structurally difficult to identify from the clue regardless of fine-tuning.

4.5.4 Overlap Between Correct Predictions

Table 4.7 shows the overlap between clues answered correctly by the baseline and fine-tuned models.

Category	**Count**	**%**
Both correct (shared)	32	7.5%
Baseline only (regression after FT)	81	19.1%
Fine-tuned only (gain after FT)	43	10.1%
Both wrong	269	63.3%
Total	**425**	**100%**

Table 4.7: Overlap between correct predictions of baseline and fine-tuned (seed 42) models on the clean subset (n=425).

Fine-tuning caused regression on 81 clues (19.1%) and improvement on only 43 clues (10.1%). The majority of clues (63.3%) were wrong for both models, which reflects the difficulty of the task. Only 7.5% of clues were solved by both models. In 77.8% of regression cases, the fine-tuned model also misidentified the wordplay type, suggesting that wordplay type misidentification is an indicator of regression.

The gain cases (43 clues) do not follow the same pattern. Reversal accounts for 11 gains, more than any other type, and fine-tuned accuracy on reversal improves by 9.4% (from 13.2% to 22.6%). This improvement is not explained by better type identification, as reversal type correctness actually drops slightly after fine-tuning (47.2% to 34.0%). The gains may instead suggest that the fine-tuned model learned patterns from the reversal training traces that happened to generalize to some test clues.

Chapter 5

Discussion

This chapter reflects critically on the results presented in Chapter 4, examines potential explanations for the observed performance drop, evaluates the validity and reliability of the findings, and situates them within the broader literature on chain-of-thought distillation and cryptic crossword solving.

5.1 Why Fine-Tuning Degraded Accuracy

The most important finding of this thesis is that fine-tuning GPT-4.1-mini on the teacher's structured reasoning traces *reduced* exact match accuracy by 9.2% compared to the zero-shot baseline. This section considers the most plausible explanations for this result.

5.1.1 Quality of the Training Signal

One of the most likely explanations is the quality of the training signal. Because the teacher was given the correct answer during trace generation (answer-informed generation), the reasoning it produced was post-hoc rationalization rather than genuine solving. As noted by Barez et al. (2025), chain-of-thought traces may not faithfully reflect the processes underlying a model's answer and may contain plausible but unsupported reasoning. Inspection of a sample of 20 accepted training traces confirmed this concern. In 9 out of 20 cases (45%), Step 4 contained errors in letter-by-letter construction despite the stated answer being correct. This suggests the teacher frequently rationalized the answer post-hoc rather than deriving it step by step.

A student model trained on such traces learns to imitate the structure and surface patterns of good reasoning without learning the underlying mechanism required to solve the task. As observed in the quality metrics, this results in a model that produces perfectly formatted five-step traces (structural completeness 5.00/5), with the answer reliably embedded in the reasoning (answer consistency 1.00). However, it still produces incorrect answers due to flawed letter-level execution. The model learned *what reasoning looks like*, not *how to reason*.

The teacher's low wordplay type correctness has important implications for the quality of the training signal. Despite achieving 45.2% accuracy, the teacher correctly identifies the wordplay type in only 17.2% of cases, which reflects the genuine difficulty of this sub-task (Sadallah et al., 2025). As shown in Section 4.5, in all 352 cases where type correctness was scored as incorrect, the teacher either named a different type or

skipped type identification altogether in Step 2. The teacher appears to solve many of these clues without explicitly naming the mechanism, reaching the correct answer through direct execution.

This does not mean that the training traces contained incorrect type labels, as the correct wordplay type was explicitly provided in the trace generation prompt. However, the traces may not have contained sufficient information about how to identify the type from the clue, both because the correct type was provided in advance and because the teacher sometimes misidentified the mechanism or omitted it altogether. Instead, they mainly showed how to explain and execute a given type. The student may therefore have learned to explain and execute a wordplay type without acquiring the reasoning needed to identify the correct type independently during evaluation.

The wordplay type correctness finding provides further evidence for this explanation. The fine-tuned model identifies fewer wordplay types correctly (0.24) than the baseline (0.41), which does not suggest that fine-tuning improved wordplay type identification. The error analysis in Section 4.5 further supports this interpretation, showing that 77.8% of regression cases also had wrong wordplay type identification.

5.1.2 Capacity Gap

Another contributing factor may be the capacity gap between teacher and student models. As Zhang et al. (2025) note, distillation gains depend on the student having sufficient representational capacity to absorb the teacher's reasoning. The gap between GPT-5.4 (45.2% zero-shot) and GPT-4.1-mini (26.6% zero-shot) on this task is significant (roughly 18%) given the difficulty of the task. This gap may be too large for effective trace-level knowledge transfer. The teacher may produce reasoning that implicitly relies on knowledge and inference capacity that the student simply does not have, which makes the traces difficult to learn from in a meaningful way.

The error analysis in Section 4.5 provides partial evidence for this explanation. Even when the fine-tuned model identifies the correct wordplay type, it produces the correct answer only 15.5% of the time. This suggests that type identification alone is insufficient and the student's execution capacity remains limited. This is consistent with the finding of Zhang et al. (2025) that distillation gains depend on the teacher-student capacity gap, with larger gaps potentially reducing the student's ability to absorb the teacher's reasoning. However, since type correctness also decreased after fine-tuning, the capacity gap alone does not fully explain the results.

5.1.3 Training Data Imbalance and Coverage

A third factor is the imbalance of the training corpus. The per-type analysis (Table 4.6) shows that anagram type correctness dropped significantly after fine-tuning (74.1% to 24.1%), despite anagram being among the most frequent training types. This suggests the model may have overfitted to surface features of training clues rather than learning a generalizable identification strategy. However, since five other types also had 1,250 training traces without showing the same drop, data quantity alone does not fully explain the pattern.

For the rarest type, alternation (396 traces, roughly a third of the 1,250 traces for abundant types), limited training data may have contributed to low type correctness, which the error analysis shows is associated with accuracy drops (Section 4.5). The

underrepresentation of alternation and deletion is unlikely to be the primary cause of the overall accuracy drop, but it likely contributes alongside the other factors discussed.

5.2 The Difference Between Trace Quality and Accuracy

The finding that structural completeness and answer-trace consistency significantly improve but accuracy declines is perhaps the most theoretically interesting result of this thesis. This shows that these trace quality dimensions do not necessarily reflect genuine reasoning ability.

This supports Lightman et al. (2024)'s argument for process supervision over outcome supervision, but also highlights a limitation: process-level metrics must measure the *validity* of reasoning steps, not just their *presence*. The metrics used in this thesis (structural completeness and answer-trace consistency) are structural rather than semantic. They verify that steps are present and that the answer appears in the reasoning, but not whether the steps are logically valid or correct at the letter level. A model that consistently produces execution steps that sound plausible but are incorrect would score highly on these metrics but perform poorly on accuracy, which is consistent with the observed results.

The format constraint introduced by fine-tuning in this thesis helps explain this discrepancy. The fine-tuned model produced responses four times shorter on average than the baseline (984 vs 4,260 characters), suggesting that it followed the five-step format more closely rather than using exploratory reasoning. This format strictness led to perfect structural completeness and answer-trace consistency, although the model still had low final answer accuracy.

An analysis of the structural dimension of regression and gain cases shows that the format constraint had mixed effects. Only one case was found where an incomplete baseline structure with a correct answer became complete but incorrect after fine-tuning. This suggests that format strictness alone does not explain the regressions. In contrast, out of the 43 clues that the fine-tuned model got right but the baseline did not, 30 (69.8%) had incomplete structure in the baseline. This suggests that the fine-tuned model's stricter format compliance may have helped on some clues where the baseline's exploratory reasoning failed to reach the correct answer. The format constraint therefore improved trace structure and helped on some clues, but was not the primary driver of the overall accuracy drop.

This finding motivates the development of more semantically grounded trace quality metrics for future work. For example, a metric that checks whether the answer can be independently verified by following the execution steps (Step 4) would detect the kind of rationalization errors seen in teacher traces. Such metrics would require either rule-based verification of letter-level operations (complex to implement for eight wordplay types) or LLM-as-a-judge evaluation (introducing concerns similar to those identified by Barez et al. (2025) about treating potentially unfaithful reasoning traces as explanations).

5.3 Reliability and Validity

5.3.1 Internal Validity

Several aspects of the experimental design strengthen the internal validity of the findings. The test set is stratified and held out from all training and development decisions. Results are replicated with three different training seeds, with a standard deviation of 0.2–0.3%, which confirms that the drop in performance is not due to randomness. Both full and clean subset versions of the results are reported, and the direction of the finding is consistent in both.

Despite these measures to ensure reliable results, some aspects of the setup may still limit internal validity. For instance, the answer extractor searches for the `ANSWER: WORD` pattern, with six additional fallbacks for alternative patterns. However, if the model produces an answer in an unexpected format not covered by the fallbacks, it will not be extracted and will be counted as incorrect.

Answer extraction failures varied in different models. The baseline (GPT-4.1-mini) had the highest failure rate with 196 out of 496 clues (39.5%). This was mainly due to reasoning loops causing response truncation before the `ANSWER:` line. The fine-tuned models had at most one failure each: seed 1 (0.0%), seed 42 (0.0%), and seed 100 (0.2%). The teacher's failures were content filter errors rather than extraction failures, and these were removed from the clean subset.

5.3.2 External Validity

The findings are limited in external generalizability in several ways. The dataset consists of English cryptic clues primarily from British newspapers, and the findings may not generalize to other cryptic crossword traditions or languages. The specific choice of teacher model (GPT-5.4) and student model (GPT-4.1-mini) introduces model-specific effects that may not be the same with other model pairs. The conservative hyperparameter configuration (learning rate multiplier 0.5, 4 epochs) was chosen to reduce the risk of catastrophic forgetting, but alternative configurations might produce different results. These remain open questions for future work.

5.3.3 Limitations of the Evaluation Metrics

As mentioned in Chapter 3, the wordplay type correctness metric uses substring matching, which may occasionally produce false positives (the type appears in a negating context) or false negatives (the teacher uses varied phrasing). As this metric is applied consistently to all models, relative comparisons are valid. However, absolute scores should be interpreted with caution.

The answer-trace consistency metric checks whether the predicted answer appears in Steps 4–5 of the trace. A model that repeats the answer multiple times throughout its trace will score highly on this metric regardless of whether its reasoning is correct. This inflates the metric's ability to distinguish between models.

The teacher model's content filtering behavior leads to 71 clues being removed from the clean subset. If these clues are systematically harder or easier than the remaining 425, the clean subset may not be representative of the full test set. Based on the per-type accuracy differences between full and clean results, the removed clues were somewhat harder on average (teacher accuracy on clean subset is 45.2% vs 38.7% on

the full test set). This would slightly inflate all clean subset results relative to the full test set.

5.4 Comparison with Related Work

The results of this thesis contrast with the positive findings of Ho et al. (2023) and Shridhar et al. (2023), who show that chain-of-thought distillation improves student accuracy on reasoning tasks. Some features of the cryptic crossword task may explain this difference.

First, cryptic crossword solving requires precise letter-level manipulation that is difficult to express and verify in natural language traces. The arithmetic and commonsense tasks studied by Ho et al. (2023) have relatively clear verification of intermediate steps. Letter-level wordplay execution does not, which makes post-hoc rationalization more likely and harder to detect.

Second, the teacher's baseline performance on this task (45.2%) is considerably lower than the teacher models used in prior distillation work, which often achieve near-ceiling performance on the tasks studied. A teacher that frequently fails may generate traces that model incorrect or incomplete reasoning patterns, which provide a noisy training signal. Zelikman et al. (2022) address a similar concern in STaR by only training on traces where the final answer is correct, which is the same answer consistency filter used in this thesis. However, this filter does not guarantee that the reasoning in those traces is logically valid.

Chapter 6

Conclusion and Future Work

6.1 Summary of Findings

This thesis investigated whether chain-of-thought reasoning trace distillation from a large language model (GPT-5.4) could improve the performance and reasoning quality of a smaller student model (GPT-4.1-mini) on the task of solving English cryptic crossword clues. A pipeline was implemented in which the teacher generated structured five-step reasoning traces for 8,863 training clues drawn from the georgeho.org dataset. The student was then fine-tuned on these traces through Azure AI Foundry. The fine-tuned student, the zero-shot baseline, and the teacher were evaluated on a stratified 496-clue test set. Exact match accuracy was used as the primary metric and three rule-based trace quality metrics as secondary measures.

The main finding is that fine-tuning on teacher reasoning traces *reduced* exact match accuracy by 9.2% on the clean subset (17.4% vs 26.6% baseline), with the result consistent across three training seeds and six out of eight wordplay types. The research question (to what extent does reasoning trace distillation improve performance and reasoning quality?) is answered primarily negatively for accuracy and positively for structural format compliance.

The most theoretically significant finding is the difference between trace quality and accuracy. The fine-tuned model achieved perfect structural completeness (5.00/5) and perfect answer-trace consistency (1.00). While it significantly exceeds the baseline on both dimensions, it scores lower on accuracy and on wordplay type identification. This confirms empirically that a model can learn to produce well-structured reasoning traces without acquiring the underlying reasoning ability those traces are meant to represent.

6.2 Contributions

The thesis makes three contributions to the literature.

First, it applies chain-of-thought reasoning trace distillation to cryptic crossword solving, a compositional reasoning task not previously addressed in the distillation literature. The negative accuracy result shows that distillation does not automatically generalize to tasks that require precise letter-level manipulation, even when the teacher model is relatively capable and the trace format is carefully designed.

Second, it implements and evaluates a multi-dimensional rule-based evaluation framework for reasoning trace quality. This framework provides an alternative to LLM-as-a-judge evaluation that is fully deterministic and reproducible. The finding

that structural trace quality and accuracy can differ has implications for how distillation pipelines should be evaluated, as improving trace formatting does not necessarily mean that distillation has succeeded.

Third, it provides a systematic analysis of distillation effects per wordplay type, which shows that the extent of performance change varies significantly by type (from -25.9% for alternation to $+3.1\%$ for reversal). This per-type analysis is absent from some distillation studies, which report only aggregate accuracy.

6.3 Future Work

Several directions follow naturally from the findings of this thesis.

Answer-blind trace generation: The most direct improvement to the pipeline would be generating teacher traces without providing the gold answer in the prompt. Answer-blind generation would force the teacher to genuinely solve the clue before explaining it, and produce traces that contain real solving strategies rather than post-hoc rationalization. The risk is a lower acceptance rate (fewer correct traces), but the quality of accepted traces would likely be higher. A similar approach to that of Zelikman et al. (2022) can be used, by training the model only on traces for which the teacher independently produced a correct answer.

Semantically grounded trace quality metrics: The structural quality metrics used in this thesis are necessary but not sufficient for evaluating reasoning validity. A step-level correctness metric that independently verifies letter-by-letter execution (e.g. checking whether the letters produced in Step 4 actually spell the stated answer) would be more informative. For some wordplay types (anagram, hidden, alternation) this is straightforwardly computable; for others (container, insertion) it requires parsing the structural decomposition from the trace text.

Different model pairs: Future work could experiment with different teacher-student pairings, where the student model is capable enough to absorb the teacher's reasoning but still meaningfully weaker to justify knowledge distillation. The finding of Zhang et al. (2025) that distillation gains depend on the capacity gap suggests that the optimal pairing is task-dependent and worth systematic investigation.

Iterative distillation: Rather than a single round of teacher trace generation and fine-tuning, iterative approaches similar to STaR (Zelikman et al., 2022) could be applied. After each fine-tuning round, the student is used to generate new traces for clues it can now solve. This would provide an expanding set of training examples grounded in the student's own reasoning.

Alternative student model families: GPT-4.1-mini is a general-purpose model capable of following instructions. Models that were specifically designed for reasoning tasks, or models with explicit chain-of-thought training in pretraining, might respond differently to being fine-tuned on reasoning traces. Similarly, applying the pipeline to open-source models (e.g. LLaMA, Mistral) would allow more detailed analysis of what changes in the model parameters during fine-tuning.

Cross-lingual extension: The georgeho.org dataset is limited to English cryptic crosswords. Dutch, German, and French cryptic crosswords exist with different conventions. Extending this pipeline to other languages would test whether the findings generalize beyond English data.

6.4 Concluding Remarks

Cryptic crossword solving is a genuinely difficult compositional reasoning task that requires natural language understanding, world knowledge, and precise letter-level manipulation. The results of this thesis suggest that chain-of-thought knowledge distillation, at least in its present form with answer-informed trace generation and the current teacher-student capacity gap, does not reliably transfer the solving ability of a capable teacher to a smaller student model. The student learns to imitate the *form* of reasoning (producing structured, format-compliant traces) without absorbing the *substance*.

This is not a failure of the distillation paradigm in general, but a finding about the conditions under which it succeeds or fails. Tasks that require precise, verifiable step-by-step manipulation may be poor candidates for standard answer-informed trace distillation. In such tasks, plausible but incorrect reasoning is easy to generate and intermediate steps are difficult to verify. Identifying these conditions, and designing distillation pipelines that address them, is an important direction for future research.

Appendix A

Data Statement

This data statement follows the recommendations of Bender and Friedman (2018) for documenting datasets used in natural language processing research.

Curation Rationale

The dataset was created by George Ho as a personal project. It is based on web-scraped data and can be used as a resource for cryptic crossword solvers and constructors. It contains a collection of clues and answers from published cryptic crosswords, amounting to over half a million instances.

Clues are scraped from three cryptic crossword blogs and a small number of digital archives. The blogs exclusively cover British newspaper crosswords (e.g. *The Times* and *The Guardian*). A clue is included if its source crossword is covered by one of the indexed blogs or archives, the blog post includes the full clues and answers, and the blog post HTML is parsable by the scraping code.

A subset of clues (from cru_cryptics and nytimes) comes from `.puz` files, which do not support rich-form clue markup, and therefore do not contain definitions or annotations.

Language Variety

The clues are in English and mainly consist of British newspaper crosswords. As a result, British English with formal register is the dominant language variety. Some clues are from North American sources (e.g. *The New York Times*), representing American English. These are not tagged separately in the dataset.

Speaker Demographic

For this dataset, 'speakers' are the creators of the published cryptic crossword puzzles who work for major British newspapers (e.g. *The Times*, *The Daily Telegraph*, *The Guardian*) and some North American publications. The creators' demographic data (age, gender, ethnicity, native language, or socioeconomic background) is not readily available, and their number is not specified in the datasheet.

Annotator Demographic

Annotating the dataset was done by the bloggers who maintain the cryptic crossword blogs from which clues were scraped. Their demographic data is not available, and may impact which clues they included or how they phrased explanations. George Ho reviewed a sample of parsed clues to check for parsing errors, but his demographic information is not mentioned either.

Speech Situation

The clues are written texts published in newspapers and online crossword publications from January 2009, with new clues added as new blog posts are published. The speech situation is formal written language, targeting puzzle-solving readers who are familiar with the cryptic crossword genre. Clues are intentionally made to be ambiguous at first reading.

Text Characteristics

The derived dataset consists of cryptic crossword clues featuring ambiguity and misleading surface reading (Efrat et al., 2021). A typical clue in the dataset can be further decomposed into three parts: (1) the letter grid (i.e. the number of letters in the final answer), (2) the definition (i.e. one or more chunks that can be used interchangeably with the final answer), and (3) the wordplay. In most cases, the wordplay part includes an indicator mapping to the wordplay type, such as *shifting* suggesting an anagram.

Moreover, as outlined in the Language Variety section, the dataset mainly consists of cryptic clues from British newspapers. As a result, the dataset is dominated by features of British cryptic crosswords, including British vocabulary, idioms, slang, and references to British culture, geography, and royalty.

Other

Human errors made by bloggers (e.g. typos, missing parts, mismatched parentheses) may occur in the dataset. Machine errors made during parsing (e.g. mismatched definitions, splitting into columns) are also possible. Some clues may be repeated if they have appeared in multiple sources.

The four columns in the dataset (annotation, puzzle_url, is_reviewed, datetime_reviewed) are not published in the public dataset due to copyright concerns, but may be made available to researchers upon request. The dataset may get updated over time, which affects the reproducibility of results. The version used in this thesis was accessed in April 2026.

The dataset is under the Open Database License, and any rights in individual contents are licensed under the Database Contents License.

Provenance Appendix

George Ho has published a datasheet for this dataset, available at: `https://cryptics.georgeho.org/datasheet`.

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